

Full model output for frequentist and Bayesian meta-analysis

This supplementary file contains the model code and output for both meta-analysis frameworks.

Frequentist meta-analysis, using {metafor}

Model code

```
rma_model <- rma(  
  yi = beta_ln,  
  sei = se_beta_ln,  
  data = data_full,  
  method = "REML"  
)
```

Output

Random-Effects Model (k = 12; tau² estimator: REML)

logLik	deviance	AIC	BIC	AICc
-24.2820	48.5640	52.5640	53.3598	54.0640

tau² (estimated amount of total heterogeneity): 3.3236 (SE = 2.0238)

tau (square root of estimated tau² value): 1.8231

I² (total heterogeneity / total variability): 84.97%

H² (total variability / sampling variability): 6.65

Test for Heterogeneity:

Q(df = 11) = 78.5314, p-val < .0001

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub
-2.4500	0.6371	-3.8453	0.0001	-3.6988	-1.2012

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Bayesian meta-analysis, using {brms}

Setting priors

```
priors <- c(
  prior(normal(-1, 2), class = Intercept),
  prior(normal(0, 2), class = sd, lb = 0)
)
```

Model code

```
m.brm <- brm(
  beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year),
  data = data,
  prior = priors,
  save_pars = save_pars(all = TRUE),
  chains = 4,
  iter = 4000
)
```

Model code: prior only (for prior predictive checks)

```
fitPrior <- brm(
  beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year),
  data = data,
  prior = priors,
  sample_prior = "only",
  chains = 4,
  iter = 4000
)
```

Output main model

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year)
Data: data (Number of observations: 12)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~author_year (Number of levels: 12)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.88	0.53	1.02	3.08	1.00	2292	3519

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-2.28	0.64	-3.54	-1.04	1.00	1796	2504

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.00	0.00	0.00	0.00	NA	NA	NA

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Output prior only sampling

Family: gaussian

Links: mu = identity; sigma = identity

Formula: beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year)

Data: data (Number of observations: 12)

Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;

total post-warmup draws = 8000

Multilevel Hyperparameters:

~author_year (Number of levels: 12)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.59	1.19	0.06	4.43	1.00	5959	3547

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-1.00	2.01	-4.90	2.99	1.00	10977	5801

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.00	0.00	0.00	0.00	NA	NA	NA

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).